

Recomputing the Godfrey–Isgur Relativized Quark Model

GIModel reproduction project

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Abstract

The relativized quark model of Godfrey and Isgur remains the standard baseline for meson spectroscopy, yet its published results cannot be regenerated from the paper alone and no open implementation of it exists. GIModel.jl recomputes the model in Julia from the published ingredients, with no refitting, following the paper’s three-stage harmonic-oscillator algorithm and carrying an independent finite-difference solver as a comparator. A companion package, GIPaper, holds the digitized reference data and the residual reports of the comparison. This report describes the computation and the architecture that keeps the model separate from its discretization, the implementation and the corrections it required, and the use of the package as an instrument for calculation and for teaching. The spectrum and its mixing stages are reproduced; complete photon-decay coverage remains open.

1 Introduction

Quantum mechanics is the rigid ground of hadron spectroscopy. A Hamiltonian gives a spectrum, its eigenvectors give matrix elements, and mixing changes both the identity of a state and the transitions it can make. Whatever else is said about a resonance — its assignment, its width, its partner — eventually reduces to those operations, and a spectroscopist who cannot carry them out for a given model is reading its predictions rather than using them.

The relativized quark model of Godfrey and Isgur (Godfrey and Isgur 1985), hereafter GI, carries out those operations across light and heavy mesons, “from the pion to the up-silon,” within a single soft-QCD Hamiltonian. One fixed set of parameters produces the masses, and afterwards the couplings, of essentially the whole meson landscape. Four decades later it is still the baseline against which new states are judged.

It is also, in practice, unavailable. The relativistic smearing, the momentum-dependent operator factors, and the isoscalar annihilation machinery are specified semiquantitatively, spread across the main text and Appendix A, and the spectra themselves are published as figures rather than tables. To use the model at all you must first rebuild it.

So I looked for a version somebody had already built. An extensive search of GitHub and the wider open-source ecosystem turns up numerical libraries — SciPy, PETSc/SLEPc, Eigen — and generic one-dimensional Schrödinger and

Salpeter solvers, and nothing that encodes the GI Hamiltonian, its mass-dependent smearing and momentum-factor operators, or its tensor, antisymmetric spin-orbit, and annihilation mixing. More broadly, there is no open implementation of constituent quark-model meson spectroscopy at all. Every reproduction therefore begins by reconstructing the Hamiltonian and its numerical conventions from the paper trail, and ends in a private directory. That absence, sitting under a field that treats the quantum-mechanical calculation as its foundation, is the practical demand this project answers.

GIModel.jl recomputes the GI spectrum from the published ingredients, taking Table II verbatim and fitting nothing. Section 2 sets out what the paper asks you to compute and how the packages are organized around it; Section 3 describes how the calculation was built, the corrections that were needed, and what the reproduction currently rests on; Section 4 describes using it, as a calculation and as teaching material; Section 5 collects what is settled and what comes next. Throughout, the 1985 paper is the authority and the code is a claim about it.

2 Computation and architecture

2.1 The model Hamiltonian

GI start from a rest-frame equation with a relativistic kinetic term,

$$H|\Psi\rangle = (H_0 + V)|\Psi\rangle = E|\Psi\rangle, \quad (1)$$

$$H_0 = \sqrt{p^2 + m_1^2} + \sqrt{p^2 + m_2^2}, \quad (2)$$

with $V = V(\mathbf{p}, \mathbf{r})$ momentum dependent and $\mathbf{p} = \mathbf{p}_1 = -\mathbf{p}_2$. In the nonrelativistic limit V reduces to the familiar sum of confinement, hyperfine, and spin-orbit terms. The spin-independent piece combines a linear Lorentz-scalar term with a Coulomb-type one-gluon exchange,

$$H_{ij}^{\text{conf}} = -\left[\frac{3}{4}c + \frac{3}{4}br - \frac{\alpha_s(r)}{r}\right] \mathbf{F}_i \cdot \mathbf{F}_j, \quad (3)$$

with $\langle \mathbf{F}_i \cdot \mathbf{F}_j \rangle = -4/3$ in a meson; the color hyperfine interaction supplies the contact and tensor pieces,

$$H_{ij}^{\text{hyp}} = -\frac{\alpha_s(r)}{m_i m_j} \left[\frac{8\pi}{3} \mathbf{S}_i \cdot \mathbf{S}_j \delta^3(\mathbf{r}) + \frac{1}{r^3} S_{ij} \right] \mathbf{F}_i \cdot \mathbf{F}_j, \quad (4)$$

and the spin-orbit interaction splits into a color-magnetic and a Thomas-precession part. A separate gluon-annihilation term H_A contributes only in self-conjugate isoscalar channels.

As GI themselves stress, Equation 4 and its spin-orbit companions are more singular than r^{-2} and are illegal operators in Equation 1 as written. The resolution is Appendix A, and Appendix A is where the reproduction work actually lives.

2.2 Relativization: smearing and momentum factors

Two things happen there. First, the relative coordinate is *smeared* over distances of order the inverse quark masses,

$$\rho_{ij}(\mathbf{r}' - \mathbf{r}) = \frac{\sigma_{ij}^3}{\pi^{3/2}} e^{-\sigma_{ij}^2(\mathbf{r}' - \mathbf{r})^2}, \quad (5)$$

with a mass-dependent width $\sigma_{ij}(\sigma_0, s)$, which tames the short-distance singularities and makes the operators legal. Second, the coefficients acquire a dependence on the quark momenta: mass factors m^{-1} are promoted to operator factors built from $E_i = (p^2 + m_i^2)^{1/2}$, so that the light-quark hyperfine term saturates to a finite $m \rightarrow 0$ limit controlled by $\langle p^{-1} \rangle$ instead of diverging.

Concretely, the central Coulomb and confinement terms become their smeared forms $\tilde{G}(r)$ and $\tilde{S}(r)$, and the Coulomb operator is dressed with a Hermitian momentum sandwich,

$$G' = A(p) \tilde{G} A(p), \quad A(p) = \left(1 + \frac{p^2}{E_1 E_2}\right)^{1/2}, \quad (6)$$

with the same construction, at the side exponents $\frac{1}{2} + \epsilon_i$ of Table II, applied to the contact, tensor, and spin-orbit kernels. Those kernels are themselves built from derivatives of the *smeared* central functions. Getting the ordering of these operations right, and keeping it right in two different numerical representations, turned out to be most of the project.

The strong coupling is a frozen sum of Gaussians,

$$\alpha_s(Q^2) = 0.25 e^{-Q^2} + 0.15 e^{-Q^2/10} + 0.20 e^{-Q^2/1000}, \quad (7)$$

with Q in GeV and $\alpha_s^{\text{crit}} = 0.60$, which becomes a coordinate-space error-function profile convenient for convolving with Equation 5. In self-conjugate isoscalars, $q\bar{q}$ annihilates through gluons ($n = 2$ for $C = +$, $n = 3$ for $C = -$), and the channel-coupling element

$$A(2S+1 L_J)_{ji} = 4\pi(2L+1) A(2S+1 L_J) \times \left[\frac{\alpha_s(M_j^2) \alpha_s(M_i^2)}{\pi^2} \right]^{n/2} \frac{S_L(\Psi_j) S_L(\Psi_i)}{m_i m_j} \quad (8)$$

depends on the wavefunction near the origin through $S_L(\Psi_i)$. That dependence makes the isoscalars the sharpest test of whether the wavefunctions are right, not merely the energies.

2.3 The paper's three-stage algorithm

The harmonic-oscillator (HO) basis was not a stylistic choice in 1985. It was the practical route through an operator

Table 1: GI Table II parameters, used verbatim. The tensor exponent is the paper's ϵ_f (code `epsilon_t`).

Quantity	Symbol	Value
Light quark mass	$\frac{1}{2}(m_u + m_d)$	220 MeV
Strange mass	m_s	419 MeV
Charm mass	m_c	1628 MeV
Bottom mass	m_b	4977 MeV
String tension	b	0.18 GeV ²
Constant	c	−253 MeV
Frozen coupling	α_s^{crit}	0.60
QCD scale	Λ	200 MeV
Smearing width	σ_0	1.80 GeV
Smearing exponent	s	1.55
Contact factor	ϵ_c	−0.168
Tensor factor	ϵ_f	+0.025
Vector spin-orbit	$\epsilon_{\text{so}(V)}$	−0.035
Scalar spin-orbit	$\epsilon_{\text{so}(S)}$	+0.055

that depends on both position and momentum: oscillator functions have tractable representations in both spaces, and Appendix A inserts intermediate states to assemble products of the corresponding matrix elements. The procedure has three stages. The relativized Hamiltonian is diagonalized in a large HO basis within fixed $|jm; ls\rangle$ sectors, the basis being enlarged until convergence. The smaller mass matrices generated by the off-diagonal tensor ($^3L_J \leftrightarrow ^3L'_J$) and antisymmetric spin-orbit ($^3L_J \leftrightarrow ^1L_J$, unequal masses only) couplings are then diagonalized in the basis of those eigenvectors. Finally, self-conjugate isoscalars acquire the annihilation matrix of Equation 8.

2.4 Inputs

The parameters are GI Table II, taken verbatim (Table~1). These constants, the α_s form of Equation 7, and the paper's reference masses are the only inputs; nothing is fitted here.

GI quote an expected accuracy of 25 MeV for light-quark mesons and 10 MeV for heavy-quark systems, dominated by neglected coupled-channel shifts and by the schematic relativization. That is the standard the reproduction is measured against, and it is worth separating early from a solver tolerance, which is a statement about arithmetic rather than about physics.

2.5 Package architecture

Two packages divide the work along the line that matters: GIModel computes, GIPaper compares. GIModel knows nothing about the 1985 figures. A meson is built from quark flavors, the levels to solve are named as explicit $n^{2S+1} L_J$ multiplets, and `compute_spectrum` returns a `Spectrum` whose states carry their contribution breakdowns and the provenance of every mixing transformation applied to them. GIPaper owns the digitized reference data, maps reference rows to mesons without fallback masses,

runs the comparison, applies the Table III annihilation prescriptions, and writes the residual reports. Nothing from the paper leaks into the calculation, so a residual can never be manufactured by the layer that reports it.

Inside GIModel the same principle separates the model from its discretization. `GIParameters` is grouped by physical aspect, one substruct per section of the parameter file, and describes the Hamiltonian only. The choice of numerical method lives on the solver — a finite-difference (FD) or an oscillator implementation of a common `RadialSolver` interface — so switching method does not rebuild the physics object, and the two methods can be run against each other on identical inputs. Wavefunctions follow the same pattern: mesh waves and oscillator waves are distinct types behind one wave interface, with the norms, overlaps, and expectation values written once against that interface. Momentum-space waves exist alongside them, which is what makes the Appendix-A momentum factors expressible in their own representation instead of through a mesh approximation.

3 Implementation

3.1 From finite differences to the oscillator basis

I did not start with the paper’s method. An FD radial calculation was the easier foothold: grid values, explicit operator matrices, and contribution plots gave something to look at directly, term by term, when nothing yet worked. That was the right first move and the wrong final answer. Once the FD spectrum stood up, its independence became the lever for going back to native HO matrix elements, and then to the paper’s own order of diagonalization and mixing. The sequence was FD to HO, with comparison at every step.

The intermediate stage was the most instructive, because it looked finished when it was not. Representing a wave in an HO expansion does not by itself reproduce the paper’s algorithm; native operator assembly, full fixed-sector diagonalization, and the later mixing stages all still had to be recovered individually. The native HO path now evaluates position- and momentum-dependent factors in their own representations following Appendix A, optimizes the oscillator scale, and enlarges the basis until every requested energy passes a stated convergence test — currently 0.1 MeV over two consecutive refinements, supplemented by wave-overlap diagnostics and wave-sensitive audits. FD remains as an independent comparator, which is a better role for it than the headline.

3.2 Corrections that taught physics

The instructive part of the work was not the code that ran. It was the code that ran *and was wrong*, and what it took to notice.

Normalization and relative phase. A diagonal Rayleigh

quotient is unchanged when its input vector is rescaled; a bare off-diagonal overlap is not. So comparing sampled FD waves with reconstructed HO waves demanded a common physical normalization, and without it perfectly matching diagonal results happily coexisted with badly wrong mixing matrix elements. The fix is recorded in commit `a987862`. The HO phase-convention correction in `9ff437b` is the same lesson from the other side: an individual eigenvector has an arbitrary phase, but every operator and every mixed-state composition must transform consistently with whatever phase was chosen. Both are statements about physics that arrived as implementation failures.

A converged approximation can still be the wrong calculation. A nonperturbative solve that was unavailable in one path had been quietly falling back to a first-order shift. Commit `eef789b` made that case fail loudly instead. The later fixed-sector work (`b9d3474`) put the full diagonal spin interactions into the Hamiltonian *before* solving it, rather than adding them afterwards. Recovering the masses was not enough: the spin-distorted wavefunctions have to reach the decay and annihilation observables too, which is exactly where a first-order shortcut stops being invisible. The completion in `78a3d76` also removed extra spin-strength bridge factors once the literal Appendix-A coefficients were correct.

Convergence certifies less than it appears to. Commit `8655d68` removed the mesh from the HO Appendix-A assembly, and `39ee9c8` made adaptive basis refinement a production requirement rather than a diagnostic. Energy convergence still had to be complemented by tests of origin-sensitive functionals and of the final mixed-state observables: an energy can be stable to 0.1 MeV while $S_L(\Psi)$ in Equation 8 is still moving. The independent FD certification in `bdd65d8` supplied a second numerical route with its own spacing and domain checks, but agreement between two routes became evidence about the physics only once each route’s representations and approximations were pinned down.

Verification infrastructure can hide the mistake. Commit `959fdb0` corrected a scorecard that was faithfully reporting the *previous* verification run, and `2fd3fd7` restored FD/HO audits that had drifted outside the gate. A correct-looking report that does not describe the code under inspection is worse than no report, and keeping the reference comparison and its generation path explicit belongs to the physics rather than to the housekeeping around it.

3.3 Agent-assisted development

Much of the work was carried out with coding agents, and the git history records its whole shape: extraction from the paper, implementation, autonomous iterations, physics audits, repairs, and eventually teaching material. The paper, the numerical checks, and my own reading supplied the criteria for accepting any of it.

The failure modes above are the reason the criteria mattered. Plausible code, passing local tests, and a reassuring residual could all coexist with a missing physical operation. What worked was the corrective loop: a discrepancy raised a question about normalization, operator staging, or provenance; the question sent me back to the paper or to an independent calculation; the answer changed both the code and the explanation of it. The commit identifiers above are navigation anchors into that record — the changes and their audits are the evidence, not the commit titles.

3.4 Provenance and current status

GIpaper keeps a manifest of paper units with links to code, tests, reference sources, and residual reports: the detailed record of what has been computed and compared. Reference data keep their origin, whether a printed table or a digitized figure. Parameter provenance is tracked with the same care, because taking the paper’s inputs, repeating a fit the paper performed, and calibrating a diagnostic control against reference targets are three different operations, and only the first two are predictions. The reports label which is which, so a control cannot quietly become evidence.

One compact example of the present state: the photon audit gives $\mu/\mu_N = +0.688$ for $\rho \rightarrow \pi\gamma$ against the printed $+0.69$, and -1.200 for $K^{*0} \rightarrow K^0\gamma$ against -1.20 , from the checked-in audit of September 7, 2026. The ρ row is also the row GI used to fit a relativization exponent, so its agreement tests the reconstructed pipeline rather than supplying an independent prediction — the distinction the provenance labels exist to preserve. Other transitions probe different flavor weights, wavefunctions, and cancellations, and are the more interesting test.

The spectrum and its mixing stages are reproduced. Complete accounting of the Table VI photon-decay rows and the characterization of their residuals are still open, and some mixing- and cancellation-sensitive quantities retain visible discrepancies. The audit files, not this report, are where that boundary is authoritative.

4 Usage

4.1 From a question to a number

The point of the architecture is that an answer can be taken apart. In the model a rest-frame mass is an eigenvalue of the Hamiltonian, whose kinetic part is Equation 2 and whose remainder is the structure of Section 2. When configurations mix, a physical state is

$$|\Psi\rangle = \sum_a c_a |\psi_a\rangle,$$

with signed coefficients and explicit flavor content, and a transition is a coherent sum,

$$\langle\Psi_f|O|\Psi_i\rangle = \sum_{a,b} c_{f,a}^* c_{i,b} \langle\psi_{f,a}|O|\psi_{i,b}\rangle.$$

Writing it out is the quickest way to see why a mixing *probability* cannot predict a decay: the relative phases can enhance or cancel. A width needs the decay operator, the normalization, and the kinematics as well.

The package exposes those stages in that order. A spectrum contains identifiable states; a state resolves into physical components; components lead to solver-native waves; waves lead to radial matrix elements; the comparison layer relates the result to a printed number. Which quantity is an input, which is computed, and which is a reference value stays visible at every step. That also makes conversational use of the model reasonable: a language model can turn a physics question into a package calculation and explain the objects that come back, while the numbers come from executing the code and carry their parameters, labels, solver settings, and convergence information with them.

4.2 Learning track and notebooks

The reproduction turned into teaching material, because the questions the code forced me to answer are the questions a student has anyway. The theory course is nine sheets: radial quantum mechanics, angular momentum and meson quantum numbers, variational and semirelativistic methods, confinement and the running coupling, relativization and smearing, spin fine structure, mixing, decay observables, and the complete spectrum algorithm. Every problem has a worked solution and a short concept check. A computational-discovery sheet starts the bridge to numerical work; the full hands-on course is still ahead.

Each piece has its own job. The paper gives the original model and its reasoning. The course slows the derivations down and makes you do them. The notebooks make the assumptions adjustable and the consequences immediate.

Pluto suits the last of those: executable cells, reactive dependencies, mathematical prose, controls, and results that redraw when a mass or a solver setting changes. Live documentation and linked docstrings make a notebook an entry point into the package rather than a demonstration beside it, and the documentation graph links the public operations explicitly, with a static link audit checking that the path from any explanation to the next numerical question exists. The adaptive HO notebook turns the convergence question of Section 3 into the lesson: it follows the oscillator scale and the basis refinements, shows the energies settling, and asks why a curve that looks flat still needs a criterion — and why energy convergence does not certify every wave-sensitive observable.

4.3 Exploring the model

The heavy-quark notebook varies a constituent heavy-quark mass continuously while the shared interaction parameters stay fixed, which connects the model’s D and B sectors, or its D_s and B_s sectors, or — varying both masses — charmonium and bottomonium. Subtracting the constituent rest masses separates the change in binding and excitation

structure from the plain rise in total mass, and hyperfine splittings, level ordering, and the response of the light degrees of freedom become things to watch rather than tabulate. The interpolation is a family of model calculations, not a physical transition between flavors.

The same access to wavefunctions supports structure pictures: radial densities, nodes, characteristic radii, angular distributions, and the components of a mixed state. These are interpretable as long as they say what they display — for a normalized reduced radial wave $|u(r)|^2 dr$ is a radial probability, while the three-dimensional density carries the angular factor as well. The ingredients for a curated meson-structure animation are in place; the animation itself is not yet.

5 Conclusion and outlook

The GI spectrum can be regenerated from the published ingredients alone, by the paper’s own three-stage algorithm, with an independent numerical route agreeing with it, and the calculation can now be inspected operator by operator instead of being reconstructed privately each time it is needed. The open work is the photon-decay coverage and the cancellation-sensitive observables that go with it.

The natural continuations are physics rather than software. Screened interactions (Li and Chao 2009) and explicit coupled-channel treatments (Barnes and Swanson 2008; Ortega et al. 2010) are the directions in which the GI baseline has mostly been extended, and each changes the model and needs its own validation. Closer to hand, the paper’s quark-emission operator can be applied directly to the calculated physical wavefunctions instead of the single-oscillator-scale approximation used for its numerical strong-decay tables, which would expose the role of radial nodes, state-dependent scales, and mixing in those amplitudes. That is the test I most want to run, because it uses the wavefunctions rather than only the eigenvalues.

Julia makes one further direction tempting. For an isolated normalized eigenstate,

$$\frac{\partial E_n}{\partial \lambda} = \left\langle \psi_n \left| \frac{\partial H}{\partial \lambda} \right| \psi_n \right\rangle$$

is an explicit starting point for sensitivity studies and fitting, and the repository already contains the numerical-type preparation and FD and fixed-scale HO derivative probes. End-to-end automatic differentiation is future work, and not trivially so: adaptive basis selection, level crossings, and derivatives of mixed eigenvectors all need care.

If the project has a moral, it is the one from Section 3: the model became clear at exactly the points where the code was confidently wrong. That is the experience the course and the notebooks are meant to make available to someone else — ask a question, get a number, take the number apart, and find out what you had assumed.

A Repository reading guide

Where the supporting material lives.

- **Learning:** `LearningTrack/README.md` and `examples/README.md` introduce the theory sheets and computational notebooks.
- **Discoverability:** `docs/discoverability_graph.md` records the linked public API; `docs/formula_map.md` connects implementation to paper equations.
- **Reproduction:** `docs/paper_manifest/` and `docs/paper_dashboard.qmd` provide the per-unit inventory. `docs/reproduction_audit.md` records the cross-cutting interpretation and remaining scope.
- **Numerical evidence:** `GIPaper/docs/residual_reports/` contains the HO convergence, FD comparator, spectrum, mixing, and decay audits. The photon numbers quoted in Section 3 come from `table_vi_photon_decays.md` there.
- **Development:** `docs/original_1985_algorithm_audit.md`, `docs/incident_followup_audit.md`, and the git history document the recovery of the paper’s algorithm and the corrections it required.
- **Differentiation:** `diff_support/implementation_status.md` distinguishes completed preparation and probes from proposed derivative capabilities.

References

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